2026 IEEE 5th World Conference on Applied Intelligence and Computing (AIC) | 979-8-3195-1898-9/26/$31.00 ©2026 IEEE | DOI: 10.1109/AIC70896.2026.11717035

PDE-Constrained DenseNet-ConvNeXt Ensemble with Total-Variation Feature Regularization for Brain Tumor Classification

Sohom Majumder
Department of Business Administration
JIS College of Engineering
Kalyani, India
sohom.majumder@jiscollege.ac.in

Rupa Paul
Department of Business Administration
JIS College of Engineering
Kalyani, India
rupa.paul@jiscollege.ac.in

Abhijit Pandit
Department of Management
Brainware University
Barasat, India
abhijitpandit1978@gmail.com

Sandip Nandi
Department of MBA
Institute of Business Management and Research
Kolkata, India
sandipnandi@rediffmail.com

Ranjana Ray
Department of Electronics and Communication Engineering
JIS College of Engineering
Kalyani, India
ranjana.ray@jiscollege.ac.in

Moumita Saha
Department of Management
Brainware University
Barasat, India
saha.moumita84@gmail.com

***Abstract* - Brain tumor classification from magnetic resonance imaging (MRI) is challenging due to acquisition noise, scanner-related artifacts, and visual similarities among tumor subtypes. This study proposes a PDE-inspired DenseNet121-ConvNeXt-Tiny ensemble with Total Variation (TV) regularization applied to intermediate feature maps. Unlike conventional regularization methods that mainly act on model parameters, the proposed TV constraint shapes spatial feature representations by encouraging smooth, noise-resistant activations while preserving discriminative tumor boundaries. The framework classifies MRI images into four categories: glioma, meningioma, pituitary tumor, and healthy. Experiments on a dataset of 7,023 brain MRI images show that the proposed TV-regularized ensemble achieves superior performance over single-backbone and non-regularized ensemble baselines, obtaining 99.24% test accuracy and 99.21% macro-F1 score. Ablation results further indicate that naïve ensemble fusion alone does not guarantee better generalization, whereas PDE-inspired feature regularization improves stability, reduces validation loss, and enhances robustness for MRI-based brain tumor classification.**

Keywords - Deep Learning; DenseNet; ConvNeXt; Total Variation Regularization; Partial differential equations

I. INTRODUCTION

Brain tumours are clinically relevant because early and accurate diagnosis directly affects treatment planning and patient outcomes. MRI is widely used because of its non-invasive nature and strong soft-tissue contrast [1]. However, MRI-based tumour classification is challenging because tumour appearance varies across patients, acquisition protocols, and scanner conditions, while different tumour subtypes may show overlapping visual features [2]. Although CNNs can learn hierarchical MRI representations for automated tumour recognition [3], high benchmark accuracy on a single dataset may not guarantee clinical reproducibility. Causes of domain shift between training and deployment settings include scanner vendor, magnetic field strength, pulse sequence, slice thickness, reconstruction pipeline, contrast protocol, patient motion, and intensity non-standardization. This is particularly important because the current dataset contains 2D JPEG images without full DICOM-level scanner metadata, making scanner-wise blind validation an important future direction.

Stronger architectures and ensembles are often used to improve performance [4], [5], but they may still learn noisy, unstable, or clinically irrelevant texture cues from non-standardized MRI data. DenseNet captures fine texture-based features through dense feature reuse [6], whereas ConvNeXt provides broader contextual representation through modern CNN design inspired by transformer-era training practices [7]. However, individual architectural strength may not be sufficient to suppress high-frequency scanner-sensitive activations. Therefore, this study proposes a PDE-constrained DenseNet121-ConvNeXt-Tiny ensemble for four-class MRI classification: glioma, meningioma, pituitary tumour, and healthy. Based on Total Variation denoising and anisotropic diffusion, the framework applies PDE-based TV regularisation directly to intermediate spatial activation maps to suppress noisy activations while preserving meaningful structural edges [8]-[10]. Its novelty lies in moving regularisation from the parameter level to the feature-representation level, making the model a boundary-preserving, noise-robust feature-regularized ensemble rather than just a larger fusion architecture.

The contributions of this work are:

- A PDE-inspired feature-regularized ensemble framework integrating DenseNet121 and ConvNeXt-Tiny for brain MRI tumour classification.

979-8-3195-1898-9/26/$31.00 ©2026 IEEE
DOI: 10.1109/AIC70896.2026.144

Authorized licensed use limited to: onsemi. Downloaded on October 07,2026 at 06:44:03 UTC from IEEE Xplore. Restrictions apply.

- Total Variation regularization on intermediate feature maps, providing an explicit variational constraint that discourages noise-driven activations during training.
- Ablation-driven validation comparing single backbones, a non-regularized ensemble, and the proposed TV-regularized ensemble under identical training and early-stopping settings.

II. Literature Review

A. Deep learning for brain MRI tumor classification

Automated brain tumour classification has transitioned from handcrafted intensity, texture, and statistical feature-based methods to modern CNN-based approaches [11]. Classical machine learning approaches were useful but required extensive feature engineering and were less reliable under variations in MRI image quality [12]. CNNs improved this process by learning features directly from input images, and architectures such as VGG and ResNet demonstrated strong ability to differentiate tumour categories through hierarchical representations [13]. More recently, ImageNet-based transfer learning has become common in medical imaging because pretrained models can improve learning efficiency and reduce the need for large labelled datasets [14].

B. DenseNet and feature reuse in medical imaging

DenseNet is widely used in medical imaging due to its dense connectivity, which enables feature reuse, improves information flow, and helps capture fine texture patterns in limited-data settings [6]. This makes it useful for brain tumour classification, where subtle texture differences are important. However, DenseNet can overfit dataset-specific or shallow visual patterns and may therefore perform poorly across different scanner and acquisition environments [15].

C. ConvNeXt and modern CNN training behaviour

ConvNeXt applies architectural and training ideas from the transformer era while retaining convolutional structure, enabling wider spatial-context representation and strong performance in medical imaging tasks [7]. However, like other advanced architectures, ConvNeXt remains sensitive to dataset noise and domain shift and may overfit high-frequency activations that perform well on one dataset but generalise poorly under different acquisition conditions [5].

D. Ensemble learning in medical image classification

Ensembling is commonly used to improve performance and reduce prediction variance by combining models that capture complementary patterns. In medical image classification, multiple backbones are often fused through averaged probabilities or logits [4]. However, ensemble gains are not guaranteed, particularly when the combined models share similar noise-related weaknesses or overfit the same dataset-specific cues [16], [17]. This motivates structured ensembles that improve not only final predictions but also the quality and robustness of learned representations.

E. Variational and PDE-inspired regularization in imaging

Variational methods and PDE-based priors have been widely used for image denoising, segmentation, and reconstruction. Total Variation regularisation is particularly useful because it promotes smooth regions while preserving important edge structures [8]. PDE-inspired deep learning methods have mainly been applied to tasks such as segmentation and reconstruction, where spatial smoothness is central [18]. However, classification models usually rely on cross-entropy loss with dropout or weight decay, leaving intermediate feature maps largely unconstrained. Applying TV regularisation directly to intermediate activations provides a structured way to create robust feature representations, particularly for MRI data affected by noise and scanner artefacts [10].

F. Positioning of the present work

The proposed method combines three research areas, which include (i) strong CNN backbones for medical image classification and (ii) ensemble fusion for complementary representation learning and (iii) variational/PDE-inspired regularization for spatial coherence. Unlike conventional ensembles that only combine outputs, this work introduces feature-level Total Variation constraints for both DenseNet and ConvNeXt activation maps, producing a regularized ensemble that is not only accurate but also more stable during validation and less susceptible to spurious high-frequency activations.

III. Research Methodology

A. Dataset Description and Pre-Processing

The dataset contains 7,023 folder-organized JPEG brain MRI images across multiple classes, posing a multi-class classification challenge due to inter-class visual similarity and intra-class variation.

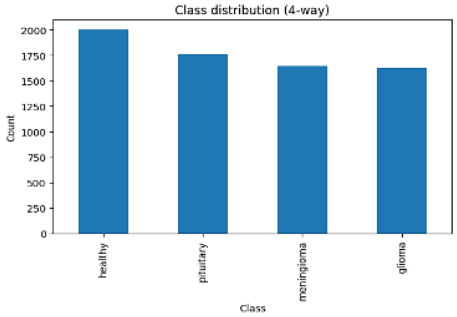


Fig. 1. (a)Class distribution of images

Fig. 1(a) illustrates the class-wise distribution of images across the four categories. Although the dataset is not perfectly balanced, the class proportions are sufficiently comparable to support reliable multiclass learning, and class imbalance is further addressed during training using categorical cross entropy loss functions. Representative MRI samples from each class are shown in Fig. 1(b).

Authorized licensed use limited to: onsemi. Downloaded on October 07,2026 at 06:44:03 UTC from IEEE Xplore. Restrictions apply.

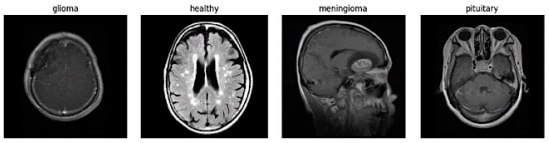


Fig. 1. (b). Representative MRI samples for glioma, healthy, meningioma, and pituitary classes

The dataset is divided into training (70%), validation (15%), and test (15%) subsets using stratified sampling to preserve class proportions. The class distribution of each image subset after partitioning into separate folders is summarized in Table I.

TABLE I. CLASS-WISE DISTRIBUTION ACROSS TRAINING, VALIDATION, AND TEST SETS

Class	Training	Validation	Test
Glioma	1135	243	243
Healthy	1400	300	300
Meningioma	1151	247	247
Pituitary	1230	263	264

All MRI images were cropped to 224 × 224 pixels, converted to three channels for ImageNet models, normalised to fixed mean and standard deviation values, and augmented only in the training set with rotation, horizontal flipping, affine transformation, and mild intensity variation; then, the dataset was split into training, validation, and test sets with stratified sampling.

B. Overall Architecture of the Proposed Framework

This is a dual-backbone ensemble architecture (i.e., the combined use of both DenseNet121 and ConvNeXt-Tiny networks), where the primary focus of most ensemble methods would be at the prediction (logit) level. The proposed approach also implements feature-space (or intermediate layer) regularization, which is based on similar concepts as Partial Differential Equations (PDEs).

The entire end-to-end processing pipeline can be broken down into the following steps:

- DenseNet and ConvNeXt backbones extract features in parallel
- Intermediate feature-map regularization via total variation (TV) constraints
- Ensemble logit fusion
- Fully end-to-end optimize using a composite loss function

A schematic overview of the proposed framework is illustrated in Fig. 2.

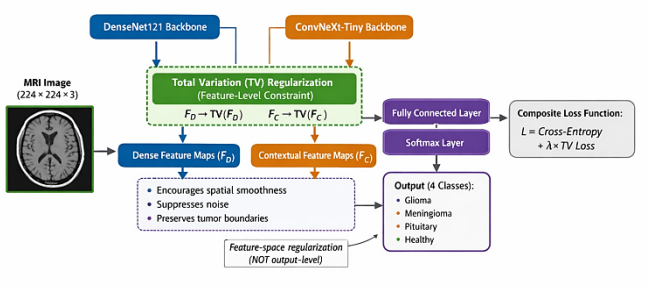


Fig. 2. Proposed PDE-Regularized DenseNet-ConvNeXt Ensemble for brain tumor MRI classification with Total Variation (TV) Regularization

C. Backbone Feature Extractors

a) DenseNet Feature Extraction: DenseNet121 was selected due to its fully dense connectivity (i.e., every feature is connected to every other feature), which allows for efficient feature reuse, thus promoting the ability to extract small-scale textural information needed for accurate diagnoses when analyzing medical images. Let $x \in \mathbb{R}^{3\times H\times W}$ denote an input MRI image. DenseNet produces an intermediate convolutional feature map

$$\mathbf{F}^{(D)} = \mathcal{F}_D(x), \mathbf{F}^{(D)} \in \mathbb{R}^{C_D\times H_D\times W_D} \qquad (1)$$

where $\mathcal{F}_D(\cdot)$ denotes the DenseNet feature extractor, and C_D, H_D, W_D represent the channel and spatial dimensions of the extracted feature map.

b) ConvNeXt Feature Extraction: ConvNeXt-Tiny is adopted as a modern convolutional architecture that incorporates training principles inspired by vision transformers while retaining convolutional inductive biases. The ConvNeXt model shows its superiority when it comes to modeling spatial dependencies and global context in a broader manner.

The ConvNeXt backbone generates an intermediate feature representation

$$\mathbf{F}^{(C)} = \mathcal{F}_C(x), \mathbf{F}^{(C)} \in \mathbb{R}^{C_C\times H_C\times W_C} \qquad (2)$$

where $\mathcal{F}_C(\cdot)$ denotes the ConvNeXt feature extractor.

D. PDE-Inspired Feature Regularization via Total Variation

a) Motivation: MRI data is noisy and can also contain scanner artifacts that are very unique to the given scanner. Standard CNNs can learn false high-frequency activations on their feature maps, leaving them with lower robustness, and performing poorly when trying to generalize. The solution is to incorporate a Total Variation (TV) regularization on

Authorized licensed use limited to: onsemi. Downloaded on October 07,2026 at 06:44:03 UTC from IEEE Xplore. Restrictions apply.

intermediate feature maps using variational processes and anisotropic diffusion equations in PDE theory.

b) Total Variation Regularization: Given a feature map $\mathbf{F} \in \mathbb{R}^{C\times H\times W}$, the anisotropic Total Variation is defined as in equation (3).

$$\mathcal{L}_{TV}(\mathbf{F}) = \sum_{c=1}^{C}\sum_{i=1}^{H-1}\sum_{j=1}^{W-1}\left(|\,\mathbf{F}_{c,i+1,j} - \mathbf{F}_{c,i,j}\,| + |\,\mathbf{F}_{c,i,j+1} - \mathbf{F}_{c,i,j}\,|\right) \quad (3)$$

The penalty placed on the feature maps due to sudden changes in value but at the same time respecting the sharp changes that indicate significant biological structure (tumour boundary).

c) Regularized Feature Learning: The constraint from the TV formulates each of the feature maps produced by DenseNet/ConvNext separately.

$$\mathcal{L}_{TV}^{(D)} = \mathcal{L}_{TV}(\mathbf{F}^{(D)}), \mathcal{L}_{TV}^{(C)} = \mathcal{L}_{TV}(\mathbf{F}^{(C)}) \quad (4)$$

This application of the TV constraints during training will help the network develop a piecewise-smooth and noise-resistant representation of features similar to the methods used in determining the solutions of Diffusion Driven partial differential equations.

E. Ensemble Fusion Strategy

Each backbone produces a class-prediction vector through a task-specific classification head

$$\mathbf{z}^{(D)} = \mathcal{H}_D(\mathbf{F}^{(D)}), \mathbf{z}^{(C)} = \mathcal{H}_C(\mathbf{F}^{(C)}) \quad (5)$$

where $\mathbf{z}^{(D)}, \mathbf{z}^{(C)} \in \mathbb{R}^K$ and $K = 4$ denotes the number of classes.

Averaging-based ensemble fusion is employed to combine predictions:

$$\mathbf{z} = \frac{1}{2}\left(\mathbf{z}^{(D)} + \mathbf{z}^{(C)}\right) \quad (6)$$

The final class probabilities are obtained using the softmax function.

F. Composite Loss Function and Optimization

The proposed model is trained end-to-end using a composite objective function that integrates classification accuracy and PDE-inspired regularization

$$\mathcal{L}_{total} = \mathcal{L}_{CE}(\mathbf{z}, y) + \lambda\left(\mathcal{L}_{TV}^{(D)} + \mathcal{L}_{TV}^{(C)}\right) \quad (7)$$

where:

$\mathcal{L}_{CE}$ denotes the categorical cross-entropy loss,

y is the ground-truth class label,

λ is a regularization hyperparameter controlling the strength of the TV constraint.

Optimization is performed using the AdamW optimizer with weight decay. Early stopping based on validation loss is employed to prevent overfitting.

G. Training Protocol and Ablation Design

To assess the contribution of each architectural component, the following ablation models are evaluated:

A1: DenseNet121 without feature regularization

A2: ConvNeXt-Tiny without feature regularization

A3: DenseNet-ConvNeXt ensemble without TV regularization

A4 (Proposed): DenseNet-ConvNeXt ensemble with TV-regularized feature learning

All models are trained under identical experimental settings to ensure a fair comparison.

H. Clinical Validity and Label Interpretation

The four diagnostic categories used in this study, glioma, meningioma, pituitary tumour and healthy, are used as labels of image-level dataset for supervised classification. The model output should therefore be viewed as a computer-aided image classification result, not as a definitive clinical diagnosis. In the clinic, the diagnosis of brain tumours is based on radiological assessment and supported by histopathological and molecular evaluation in accordance with established medical standards, such as the WHO Classification of Tumours of the Central Nervous System. The current study is not intended to replace such clinical standards, but rather to develop a computational screening framework that could aid radiologists by enhancing the robustness of features under noisy MRI conditions [19-21].

IV. Results and Discussion

A. Experimental outcomes and overall performance

Table II compares the proposed PDE/TV-regularized ensemble with CNN, non-regularized ensemble and transformer-based baselines. Although Swin Transformer-Tiny and the hybrid CNN-Transformer model show competitive performance, A4 achieves the highest test accuracy and macro-F1, indicating that feature-level TV regularisation enhances generalisation by decreasing noise-sensitive activations in MRI-based brain tumour classification.

TABLE II. Performance Comparison with CNN, Ensemble, and Transformer-Based Models

Model	Best Epoch	Test Accuracy	Macro-F1
A4_Ensemble_Avg_WithTV (Proposed)	**12**	**0.992**	**0.991**

Authorized licensed use limited to: onsemi. Downloaded on October 07,2026 at 06:44:03 UTC from IEEE Xplore. Restrictions apply.

A1_DenseNet121	8	0.990	0.984
Hybrid CNN-Transformer	13	0.989	0.987
Swin Transformer-Tiny	14	0.987	0.985
ViT-B/16	15	0.986	0.984
A3_Ensemble_Avg_NoTV	11	0.985	0.982
A2_ConvNeXtTiny	12	0.982	0.981

B. Ablation and Generalization Analysis

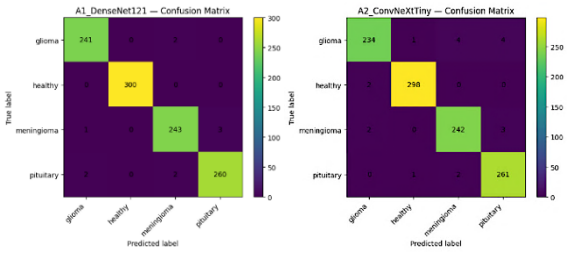
The ablation results (see Table III) show that DenseNet121 (A1) performs better than ConvNeXt-Tiny (A2), indicating stronger feature extraction for this dataset. However, the non-regularized ensemble (A3) does not improve over A1, suggesting that simple logit averaging may preserve nuisance-driven or noise-sensitive activations. In contrast, the proposed TV-regularized ensemble (A4) achieves the best overall performance by imposing spatial smoothness on intermediate feature maps, suppressing high-frequency scanner-sensitive activations while preserving discriminative tumor boundaries. Its low validation loss of 0.0328 and high validation accuracy of 0.9915, closely aligned with the final training accuracy of 0.9896, indicate stable convergence and improved generalization. Compared with A1 and A2, which show higher validation loss or larger train-validation gaps, A4 demonstrates that PDE/TV-based feature regularization acts as an effective inductive constraint rather than a superficial ensemble enhancement.

TABLE III. ABLATION RESULTS

Model	Final train loss	Final val loss	Final train acc	Final val acc	Test acc	Macro f1
A4_Ensemble_Avg_WithTV	0.029	0.032	0.989	0.991	0.992	0.991
A1_DenseNet121	0.040	0.063	0.984	0.981	0.990	0.984
A3_Ensemble_Avg_NoTV	0.036	0.046	0.990	0.983	0.985	0.982
A2_ConvNeXtTiny	0.042	0.083	0.986	0.971	0.982	0.981

C. Class-wise error patterns and confusion-matrix analysis

Fig. 3. shows that A4 has stronger diagonal dominance and fewer inter-class errors than the non-regularized ensemble, with most remaining misclassifications occurring between glioma and meningioma due to overlapping 2D MRI texture patterns.



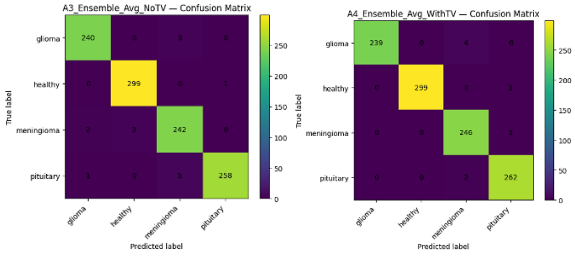


Fig. 3. Confusion Matrices for all the Models

The healthy class is consistently easiest to classify, while errors mainly occur among tumor subtypes due to overlapping features; notably, A4 reduces these misclassifications compared to A3, supporting that TV regularization enhances stable, boundary-aware representations over noise-driven decisions.

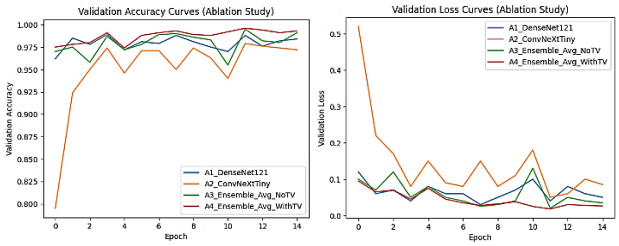


Fig. 4. Validation Accuracy and Loss curves of the Models

Fig. 4. shows that A4 achieves the most stable convergence with the lowest loss variation, while A2 shows greater overfitting and A3 remains less stable despite ensemble fusion.

TABLE IV. STATISTICAL SIGNIFICANCE ANALYSIS OF THE PROPOSED A4 MODEL

Comparison	Δ Accuracy	Δ Macro-F1	McNemar χ^2	p-value	95% CI for Δ Accuracy	95 % CI for Δ Macro-F1	Result
A4 vs A1_DenseNet121	+0.002	+0.007	0.250	0.617	[-0.002, 0.006]	[0.001, 0.013]	Not significant
A4 vs A3_Ensemble_Avg_NoTV	+0.007	+0.009	5.143	0.023	[0.001, 0.013]	[0.003, 0.015]	Significant
A4 vs A2_ConvNeXtTiny	+0.010	+0.010	7.692	0.006	[0.004, 0.017]	[0.004, 0.017]	Significant
A4 vs ViT-B/16	+0.006	+0.007	4.267	0.039	[0.001, 0.012]	[0.002, 0.013]	Significant
A4 vs Swin Transformer-Tiny	+0.005	+0.006	3.938	0.047	[0.000,	[0.001,	Significant

Authorized licensed use limited to: onsemi. Downloaded on October 07,2026 at 06:44:03 UTC from IEEE Xplore. Restrictions apply.

					0.011]	0.012]	
A4 vs Hybrid CNN-Transformer	+0.003	+0.004	1.800	0.180	[-0.001, 0.008]	[-0.001, 0.010]	Not significant

Table IV shows the paired statistical comparison of the proposed A4 model versus the competing baselines. McNemar's test was used since all models were evaluated on the same test samples and therefore their classification errors were paired and not independent. The results show that A4 significantly outperforms the non-regularized ensemble, ConvNeXt-Tiny, ViT-B/16, and Swin Transformer-Tiny models in a statistically significant manner. However, the improvement over DenseNet121 and the Hybrid CNN-Transformer baseline is not statistically significant, but A4 still achieves the best numerical accuracy and macro-F1 score.

D. Computational cost and practical considerations

Runtime comparison indicates that the ensemble variants are substantially more expensive than single-backbone models. DenseNet-only completes in roughly 842 s, while ConvNeXt-only requires about 1657 s. The ensemble models take approximately 2526-2532 s, reflecting the doubled forward/backward passes and the overhead from additional feature handling and regularization. The proposed model demonstrates better accuracy through its fundamental research method which proves that TV regularization functions as an essential variational constraint instead of a temporary solution.

While A3 and A4 have similar foundations and share commonalities from an architectural standpoint, A4 provides a slightly higher degree of regularization through piece-wise smooth properties that align with the underlying anatomical structure. This allows for an improvement in validation stability and a decrease in misclassification through its function as a clinically relevant regularized representation learning model, compared to simply being a larger model.

V. Conclusion and Future Work

In this work, we propose a PDE-constrained DenseNet-ConvNeXt ensemble with Total Variation (TV) regularisation for brain tumour classification, which can provide spatial smooth, noise-resistant and boundary-aware feature representations. Experimental results indicate that direct ensemble fusion does not necessarily yield improvement in generalisation, whereas the presented TV-regularized model (A4) achieves the highest accuracy, macro-F1 score, validation stability and the overall generalisation capability by suppressing the noise-induced and high-frequency activations in intermediate feature maps. The framework is modular and can be integrated into existing computer aided diagnosis pipelines based on MRI as a supporting classification tool. Future work will focus on generalising the model to three-dimensional MRI volumes, evaluating the robustness of the model on multi-institutional and scanner-wise datasets, and conducting a detailed sensitivity analysis of the TV regularisation parameter to evaluate the influence of different regularisation strengths on accuracy, macro-F1 score, validation loss, smoothness of feature-map, and preservation of discriminative tumour boundaries.

References

[1] K. S. Kumar, A. Bansal, and N. P. Singh, "Brain tumor classification using deep learning techniques," in *Machine Learning, Image Processing, Network Security and Data Sciences (MIND 2022)*, Commun. Comput. Inf. Sci., vol. 1763. Cham, Switzerland: Springer, 2023, pp. 68-81, doi: 10.1007/978-3-031-24367-7_6.

[2] B. M. Ali, T. H. Alfallj, M. M. Alqathaniyyin, M. M. Ashoor, R. M. Alqurashi, A. Q. Alrashidi, T. Y. Futayni, R. M. Alhussain, A. S. Alzhrani, and R. A. Alobidi, "Neuroimaging techniques in the early detection of brain tumors," *J. Healthcare Sci.*, vol. 4, no. 10, Art. no. JOHS2024000910, 2024, doi: 10.52533/JOHS.2024.41012.

[3] S. Muthuraj, R. D. Chintamani, Y. S. Deshmukh, and P. R. Mutkule, "Neural networks in neuroimaging: A critical analysis of deep learning techniques for brain tumor prediction," *J. Electr. Syst.*, vol. 20, no. 2, pp. 1617-1636, 2024, doi: 10.52783/jes.1468.

[4] K. V. Paluri and A. Gupta, "ETLoViT: An acne diagnose approach using vision-transformers and model ensembling," *Eng. Res. Express*, vol. 6, no. 3, Art. no. 035242, 2024.

[5] A. Y. Jaffar, "Combining local and global feature extraction for brain tumor classification: A vision transformer and iResNet hybrid model," *Eng., Technol. Appl. Sci. Res.*, vol. 14, no. 5, pp. 17011-17018, Oct. 2024, doi: 10.48084/etasr.8271.

[6] A. D. Swaminathan, A. Begum, K. Ramamoorthy, R. N. Nadarajan, S. Krishnamoorthy, P. K. Balachandran, and S. Kannan, "Hybrid deep learning for brain tumor detection: Combining DenseNet and custom CNN for enhanced accuracy," *Biomed. Eng. Comput. Biol.*, vol. 16, 2025, doi: 10.1177/11795972251395954.

[7] W. Ilham and A. Ahmad, "A comprehensive review of ConvNeXt architecture in image classification: Performance, applications, and prospects," *Int. J. Adv. Comput. Inform.*, vol. 2, no. 2, pp. 108-114, 2026, doi: 10.71129/ijaci.v2i2.pp108-114.

[8] Y. Li, D. He, and Y. Li, "Underwater visual image denoising based on total variation method," in *Proc. SPIE 12565, Conf. Infrared, Millimeter, Terahertz Waves Appl. (IMT 2022)*, 2023, Art. no. 125652R, doi: 10.1117/12.2662724.

[9] R. S. and S. C. K., "Image denoising using deep learning and wavelet transform," *Int. J. Sci. Technol.*, vol. 16, no. 3, Aug. 2025, doi: 10.71097/IJSAT.v16.i3.7554.

[10] N. Wen, Q. Liu, and L. Xiao, "Perona-Malik diffusion driven CNN for supervised classification of hyperspectral images," in *Proc. 2020 IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Waikoloa, HI, USA, 2020, pp. 850-853, doi: 10.1109/IGARSS39084.2020.9323535.

[11] A. Chakravarthy R., "Optimized brain tumor detection using Python based image processing," *Int. J. Sci. Res. Eng. Manage.*, vol. 9, no. 5, pp. 1-9, 2025, doi: 10.55041/IJSREM47854.

[12] H. Kibriya, M. Masood, M. Nawaz, R. Rafique, and S. Rehman, "Multiclass brain tumor classification using convolutional neural network and support vector machine," in *Proc. 2021 Mohammad Ali Jinnah Univ. Int. Conf. Comput. (MAJICC)*, 2021, pp. 1-4, doi: 10.1109/MAJICC53071.2021.9526262.

[13] R. R. Ali, N. M. Yaacob, M. H. Alqaryouti, A. E. Sadeq, M. Doheir, M. Iqtait, E. H. Rachmawanto, C. A. Sari, and S. S. Yaacob, "Learning architecture for brain tumor classification based on deep convolutional neural network: Classic and ResNet50," *Diagnostics*, vol. 15, no. 5, Art. no. 624, 2025, doi: 10.3390/diagnostics15050624.

Authorized licensed use limited to: onsemi. Downloaded on October 07,2026 at 06:44:03 UTC from IEEE Xplore. Restrictions apply.

[14] A. Kanavos, O. Papadimitriou, G. Vonitsanos, M. Maragoudakis, and P. Mylonas, "Enhanced brain tumor classification with convolutional neural networks," in *Advances in Experimental Medicine and Biology*, 2026, pp. 529-536, doi: 10.1007/978-3-032-03398-7_48.

[15] F. Khader, J. Schock, D. Truhn, F. Morsbach, and C. Haarburger, "Adaptive preprocessing for generalization in cardiac MR image segmentation," in *Statistical Atlases and Computational Models of the Heart: M&Ms and EMIDEC Challenges*, Lecture Notes in Computer Science, vol. 12592. Cham, Switzerland: Springer, 2021, pp. 269-276, doi: 10.1007/978-3-030-68107-4_27.

[16] Y. Shi, Y. Zhang, and S. Wang, "Competitive ensembling teacher-student framework for semi-supervised left atrium MRI segmentation," in *Proc. 2023 IEEE Int. Conf. Bioinformatics Biomed. (BIBM)*, Istanbul, Turkey, 2023, pp. 2231-2234, doi: 10.1109/BIBM58861.2023.10385958.

[17] W. Hang, Y. Huang, S. Liang, B. Lei, K.-S. Choi, and J. Qin, "Reliability-aware contrastive self-ensembling for semi-supervised medical image classification," in *Medical Image Computing and Computer Assisted Intervention-MICCAI 2022*, Lecture Notes in Computer Science, vol. 13431. Cham, Switzerland: Springer, 2022, pp. 754-763, doi: 10.1007/978-3-031-16431-6_71.

[18] Z. Duan, M. R. Min, L. E. Li, M. Cai, Y. Xu, and B. Ni, "Disentangled deep autoencoding regularization for robust image classification," arXiv:1902.11134, 2019, doi: 10.48550/arXiv.1902.11134.

[19] Ellingson, B.M., Bendszus, M., Boxerman, J., Barboriak, D., Erickson, B.J., Smits, M., Nelson, S.J., Gerstner, E., Alexander, B., Goldmacher, G. and Wick, W., 2015. Consensus recommendations for a standardized brain tumor imaging protocol in clinical trials. Neuro-oncology, 17(9), pp.1188-1198..

[20] D. N. Louis, A. Perry, P. Wesseling, D. J. Brat, I. A. Cree, I. Figarella-Branger, C. Hawkins, H. K. Ng, S. M. Pfister, G. Reifenberger, R. Soffietti, A. von Deimling, and D. W. Ellison, "The 2021 WHO classification of tumors of the central nervous system: A summary," *Neuro-Oncology*, vol. 23, no. 8, pp. 1231-1251, Aug. 2021, doi: 10.1093/neuonc/noab106.

[21] Baid, U., Ghodasara, S., Mohan, S., Bilello, M., Calabrese, E., Colak, E., Farahani, K., Kalpathy-Cramer, J., Kitamura, F.C., Pati, S. and Prevedello, L.M., 2021. The rsna-asnr-miccai brats 2021 benchmark on brain tumor segmentation and radiogenomic classification. arXiv preprint arXiv:2107.02314..

Authorized licensed use limited to: onsemi. Downloaded on October 07,2026 at 06:44:03 UTC from IEEE Xplore. Restrictions apply.